

The fundamentals of data visualization in R

Day 3

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1 Quarto

What we normally do

1. Use Office or $\text{\LaTeX}$ to write up a paper, a presentation, etc.
2. Use SPSS, SAS, Python, Julia, R to run our analysis
3. Use EndNote, Mendeley, etc. or $\text{\BibTeX}$ for references

This is probably the workflow of most people in linguistics. There are **two** modern alternatives here:

A. Use Typst instead of $\text{\LaTeX}$: it's faster, lighter, more intuitive, and more powerful (replaces 1)

B. Use Quarto, which combines 1–3 above (or Typst) in a **single file** based on Markdown: `Qmd`

Quarto: Today we combine everything we've seen into a single Quarto file. Plus, we'll add some linguistics to it using two specific packages that help us visualize representations.

Together or separate?

Some people prefer to separate the analysis from the text. Different styles (subjective)

Together

- A single file with code, references, text
- Intuitive and easy to reproduce/update
- Usually multiple scripts in the background
- Easy to automate

Separate

- Different files, but simpler syntax
- More control (directly set Typst or $\text{\LaTeX}$)
- Analysis in scripts, text in a dedicated file
- Can still be automated, but less straightforward

Why Quarto?

The goal is not a new editor; the goal is a reproducible document.



Write

paper, slides, report



Render

one source file



Analyze

R scripts and data files

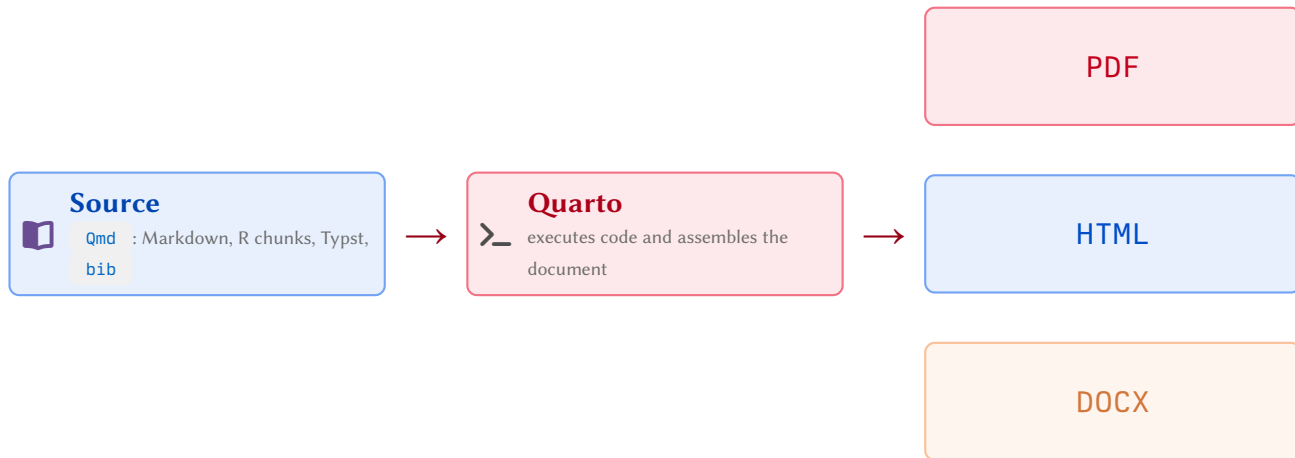


Share

PDF, HTML, Word, slides

The Quarto pipeline

A `.qmd` document is a recipe: prose + code + metadata + citations.



Source file = document; output files = products we regenerate.

What goes in a **Qmd** file?



YAML

title, author, bibliography, output format



Markdown

sections, paragraphs, lists, links



Code chunks

R code that creates numbers, tables and figures



References

labels, cross-references and **bib** entries

```
---  
title: "A paper in linguistics"  
format: typst  
bibliography: references.bib  
---
```

Small document, big change

The sentence and the result stay connected.

```
The mean reaction time is  
`r mean(d$RT) > round(2)`  
ms.  
  
```{r}  
ggplot(d, aes(Affix, RT)) +
 stat_summary()
```
```

- the number is computed from the same data as the figure
- the figure is regenerated when the code changes
- cross-references and citations are checked during rendering
- no copy-paste step between analysis and writing

Where Typst enters

Quarto controls the workflow; Typst controls PDF typesetting.



- Use `format: typst` when you want Quarto to create a PDF through Typst
- Typst is lighter than $\text{\LaTeX}$ for many day-to-day documents
- Typst is also programmable, so it can draw linguistic representations directly

For us: Quarto handles analysis and rendering. Typst gives us high-quality PDF output and domain-specific linguistic figures.

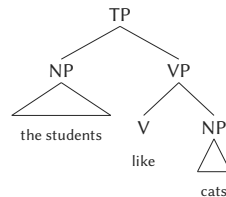
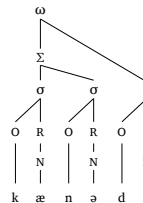
Typst inside Quarto

Typst code can be embedded in a `Qmd` document when the format is Typst.

```
```{=typst}
#import "@preview/phonokit:0.5.9": *
#import "@preview/synkit:0.0.3": *
```

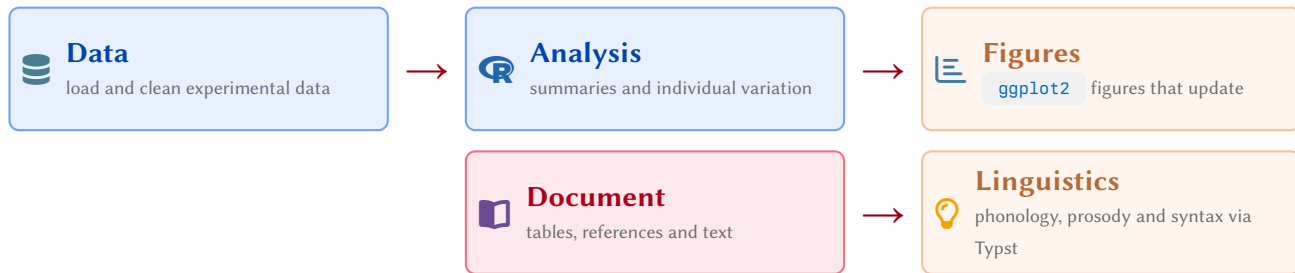
`#word("('k \\ae.n@).d@")`{=typst}

`#tree("[TP [NP the students]
      [VP like cats]]")`{=typst}
```



What we will build today

This connects the final day to the course's bigger goal: publication-ready data visualization.



(Wickham, 2016; Wickham et al., 2019)

Habits that make this work

- Treat the `Qmd` source as the **real document**
- Give important outputs labels: `fig- ...` , `tbl- ...`
- Keep figure code close to the text that interprets it
- Use `bib` entries early, even for drafts
- Render often: small errors are easier to fix than a broken final document
- Commit or save working versions before major edits

References

- Wickham, H. (2016). *ggplot2: elegant graphics for data analysis* (2nd ed.). Springer. <https://doi.org/10.1007/978-3-319-24277-4>
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Golemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., ... Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686. <https://doi.org/10.21105/joss.01686>